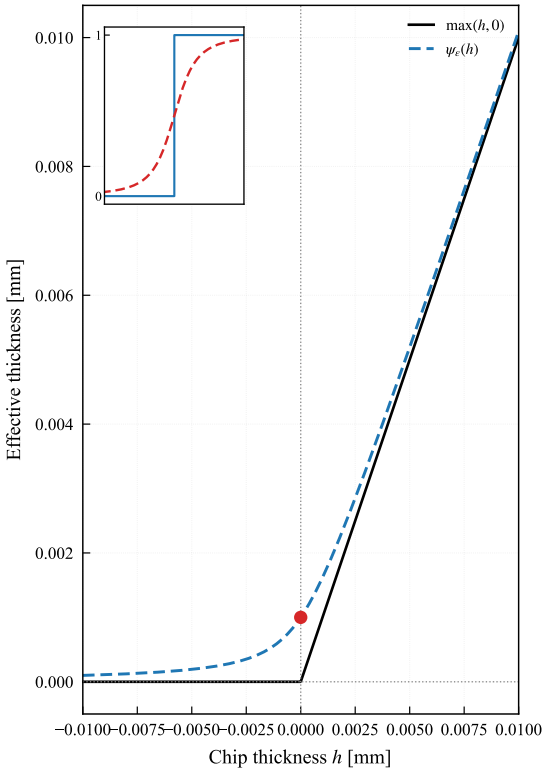
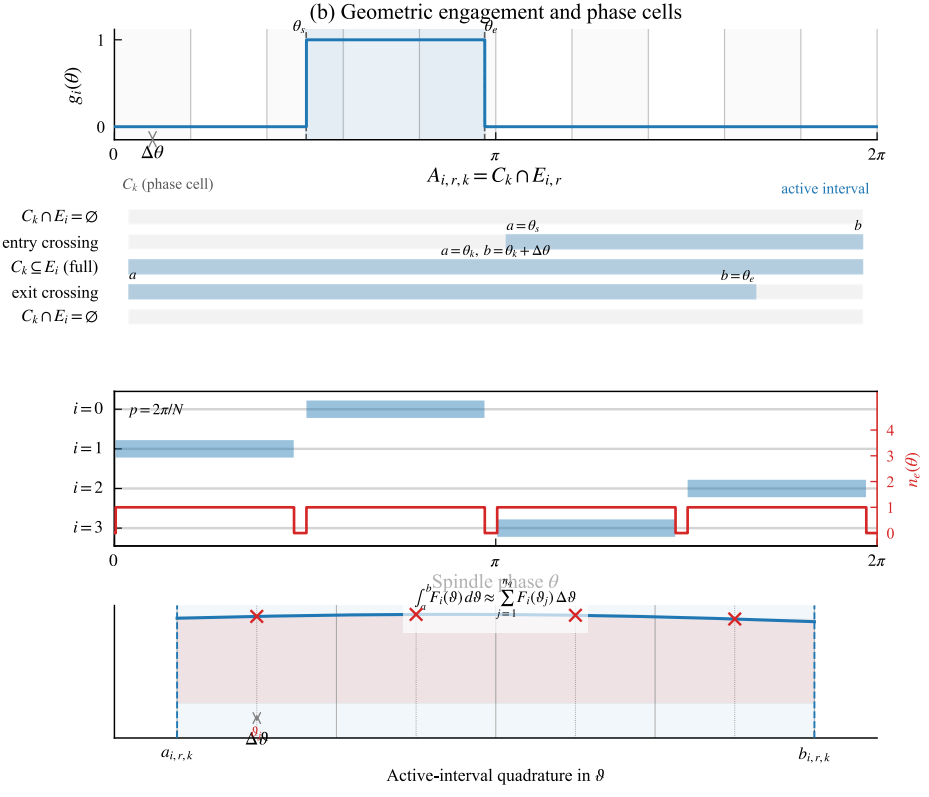

(a) Chip-loss regularisation
max(h, 0)
ψε(h)
Effective thickness [mm]
Chip thickness h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100


(b) Geometric engagement and phase cells
gi(θ)
θs
θe
Δθ
0
π
2π
Ck (phase cell)
Ai,r,k = Ck ∩ Ei,r
active interval
Ck ∩ Ei = ∅
entry crossing
Ck ⊆ Ei (full)
exit crossing
Ck ∩ Ei = ∅
a = θs
b
a = θs, b = θk + Δθ
a
b = θe
i = 0
i = 1
i = 2
i = 3
p = 2π/N
nε(θ)
Spindle phase θ
∫ Fi(ϑ) dϑ ≈ Σ Fi(ϑj) Δϑ
ai,r,k
bi,r,k
Active-interval quadrature in ϑ